

VERACISM System Testing Documentation

System: VERACISM – Verifiable Academic Records for International School Manila
Institution: International School Manila (ISM)
Test Execution Tool: Newman (Postman CLI Runner)
Test Environment: Dockerized full-stack (Angular frontend, .NET 8 backend, ClamAV, Lighthouse Encryption Service)

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1. Overview of Testing Strategy

VERACISM (Verifiable Academic Records for International School Manila) underwent two distinct automated test suites to validate the correctness, security, and integrity of the document upload and verification pipeline.

Suite	Report Title	Executed	Tool
Integration	VERACISM API v1 Integration Tests	2026-03-30	Newman
Functional	VERACISM Functional Test Suite	2026-03-31	Newman

Integration tests target the API layer directly via API key authentication, covering all core endpoints (upload/scan, report retrieval, public verification, file download, and storage status) with both positive and negative cases.

Functional tests simulate real user journeys aligned to defined user roles — IT Office staff, Registrar staff, Students/external verifiers — and an end-to-end upload-to-verify journey. Tests use Bearer token (Auth0 JWT) authentication to reflect production authorization flows.

All test assertions were run against a live instance of the system. All **182 assertions** across **41 test suites** passed with **zero failures and zero errors**.

2. Testing Methodology

2.1 Testing Approach

VERACISM was tested using **black-box API testing**, where tests interact exclusively through the system's published HTTP endpoints without knowledge of internal implementation details. This approach mirrors how real clients (frontend applications, third-party integrators, and external verifiers) interact with the system.

Two complementary test layers were employed:

Layer	Focus	Auth Method	Scope
Integration Testing	API contract correctness, error handling, response schema validation	API Key (X-Api-Key header)	All API endpoints, both happy-path and negative cases
Functional Testing	End-to-end user journey validation aligned to roles	Bearer Token (Auth0 JWT)	Role-based workflows: IT Office, Registrar, Student/External

2.2 Test Design Principles

Equivalence Partitioning was applied to group inputs into valid and invalid classes. For each endpoint, at minimum one positive case (valid input, expected success) and one or more negative cases (invalid credentials, missing fields, unknown identifiers) were defined.

Boundary Value Analysis was used for fields with format constraints — for example, SHA-256 hash outputs were asserted to be exactly 64 hexadecimal characters, and response times were bounded at 30 seconds.

Role-Based Testing structured the functional suite around the three primary user roles defined in the system:

- **IT Office Staff** — authenticated API upload and metadata retrieval
- **Registrar Staff** — document upload with student association, controlled download
- **Students / External Verifiers** — unauthenticated public CID verification

Negative Testing was given equal weight alongside positive cases. Every authenticated endpoint was tested with missing and invalid credentials. Every lookup endpoint was tested with unknown and malformed identifiers.

2.3 Test Execution Environment

Tests were executed against a **staging environment** running the full VERACISM stack via Docker Compose:

Component	Technology	Role
Frontend	Angular 20	Web UI (not directly tested in API suites)
Backend API	.NET 8 (ASP.NET Core)	Primary test target
Database	SQL Server	Persistent storage for reports and users
Threat Scanner	ClamAV + YARA	Malware and signature detection
Decentralized Storage	Lighthouse (IPFS/Filecoin) via Node.js proxy	Document anchoring and retrieval
Identity Provider	Auth0	JWT Bearer token issuance

API Base URL: <https://sb-api.ismanila.org/Veracism>

Frontend URL: <https://veracism.pages.dev>

2.4 Tooling

Tool	Purpose
Postman	Test collection authoring and environment variable management
Newman	CLI-based headless test runner for automated execution
Newman HTML Reporter	Human-readable HTML test reports (report.html , functional-report.html)
JUnit XML Reporter	Machine-readable results for CI/CD integration (results.xml , functional-results.xml)

2.5 Pass / Fail Criteria

A test case is considered **passed** when all of the following hold:

- The HTTP response status code matches the expected value
- All response body assertions (field presence, type, format, value) are satisfied
- Response time is under 30 seconds (global assertion applied to every request)

A test run is considered **successful** only if **all assertions pass with zero failures and zero errors**.

2.6 Traceability

Each test case ID (e.g., [TC-01](#), [TC-ITO-001](#), [TC-E2E-001](#)) maps directly to a requirement or user story in the functional specification. The functional suite test IDs follow a role-prefix convention:

Prefix	Role / Area
TC-ITO-	IT Office staff workflows
TC-REG-	Registrar staff workflows
TC-STU-	Student-facing workflows
TC-EXT-	External verifier workflows
TC-E2E-	End-to-end cross-role journeys
TC-## (numeric)	Integration-layer API contract tests

Appendix A — Integration Test Suite Results

Suite Name: VERACISM API v1 Integration Tests
Execution Date: 2026-03-30
Total Test Suites (Collections): 22
Total Assertions: 93
Failures: 0 | **Errors:** 0
Total Execution Time: 83.728 seconds

A.1 — Group 1: Scan Upload (POST /api/scan)

TC-01 — Upload Safe PDF

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	20.446
2	Status 200	✓ PASS	20.446
3	Has requestId (GUID or 32-hex)	✓ PASS	20.446
4	Results array has at least 1 item	✓ PASS	20.446
5	sha256 is 64-char hex	✓ PASS	20.446
6	threatLevel is valid	✓ PASS	20.446
7	reportId is a number	✓ PASS	20.446

TC-02 — Upload with studentNumber

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	17.321
2	Status 200	✓ PASS	17.321
3	studentId is non-null when studentNumber provided	✓ PASS	17.321

TC-03 — Upload with documentLabel and parentReportId (Versioning)

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	17.392
2	Status 200	✓ PASS	17.392
3	versionNumber is at least 2	✓ PASS	17.392
4	documentLabel present in response	✓ PASS	17.392

TC-04 — No FILES field — expect 400

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.054
2	Status 400	✓ PASS	0.054
3	Error is 'no files'	✓ PASS	0.054

TC-05 — Invalid API Key — expect 401 or 403

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.134
2	Status 401 or 403 for invalid API key	✓ PASS	0.134

TC-06 — Missing X-API-Key Header — expect 401 or 403

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.050
2	Status 401 or 403 when no API key	✓ PASS	0.050

TC-07 — Upload TXT File

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	16.888
2	Status 200	✓ PASS	16.888
3	Results array present for txt upload	✓ PASS	16.888
4	sha256 present in txt result	✓ PASS	16.888

TC-07A — Upload JPG Flagged by YARA — not uploaded to Lighthouse

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	4.734
2	Status 200	✓ PASS	4.734
3	Has requestId	✓ PASS	4.734
4	Results array has at least 1 item	✓ PASS	4.734
5	Returned file is MyPHOTO_2.jpg	✓ PASS	4.734
6	reportId is a number	✓ PASS	4.734
7	sha256 is 64-char hex	✓ PASS	4.734
8	ClamAV says not infected	✓ PASS	4.734
9	YARA detected a match	✓ PASS	4.734
10	YARA rule includes image webshell signature	✓ PASS	4.734
11	Threat level is not SAFE	✓ PASS	4.734
12	Lighthouse upload is blocked	✓ PASS	4.734

TC-08 — Get Report by Valid CID

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.147
2	Status 200	✓ PASS	0.147
3	CID in response matches environment	✓ PASS	0.147
4	sha256 is 64-char hex	✓ PASS	0.147
5	filename is a non-empty string	✓ PASS	0.147
6	status is 'complete'	✓ PASS	0.147
7	verificationUrl is non-empty string	✓ PASS	0.147
8	threatLevel is valid enum	✓ PASS	0.147
9	uploadedAt is a date string	✓ PASS	0.147

TC-09 — Get Report Unknown CID — expect 404

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.054
2	Status 404	✓ PASS	0.054
3	Error is 'Report not found'	✓ PASS	0.054

TC-10 — Get Report No API Key — expect 401 or 403

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.048
2	Status 401 or 403 with no API key	✓ PASS	0.048

TC-11 — Tenant Isolation — Unknown CID returns 404, not 403

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.053
2	Status 404 for non-owned CID (tenant enumeration hidden)	✓ PASS	0.053
3	Response is not 403 (no tenant leakage)	✓ PASS	0.053

A.3 — Group 3: Verify — Public Endpoint (GET /api/verify/:cid)

TC-12 — Verify Valid CID — verified: true

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.066
2	Status 200	✓ PASS	0.066
3	verified is true	✓ PASS	0.066
4	cid matches	✓ PASS	0.066
5	sha256 is 64-char hex	✓ PASS	0.066
6	uploadedAt is present	✓ PASS	0.066
7	threatLevel is valid	✓ PASS	0.066

TC-13 — Verify Unknown CID — verified: false with message

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.055
2	Status 200 (not 404 — verify returns 200 for unknown CIDs)	✓ PASS	0.055
3	verified is false	✓ PASS	0.055
4	message is present	✓ PASS	0.055

TC-14 — Verify Garbage CID — verified: false with message

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.053
2	Status 200	✓ PASS	0.053
3	verified is false for garbage CID	✓ PASS	0.053
4	message is present	✓ PASS	0.053

TC-15 — Verify Public — No API Key Must Still Return 200

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.056
2	Status 200 without API key (public endpoint)	✓ PASS	0.056
3	verified is true even without auth	✓ PASS	0.056

A.4 — Group 4: Download ([GET /api/verify/download/:cid](#))

TC-16 — Download Valid File — Binary Response

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	5.841
2	Status 200	✓ PASS	5.841
3	Content-Disposition is attachment	✓ PASS	5.841
4	Content-Type header is present	✓ PASS	5.841
5	Response body is non-empty	✓ PASS	5.841

TC-17 — Download Unknown CID — expect 404

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.054
2	Status 404	✓ PASS	0.054
3	Error is 'Report not found'	✓ PASS	0.054

TC-18 — Download No API Key — expect 401 or 403

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.119
2	Status 401 or 403 when no API key	✓ PASS	0.119

A.5 — Group 5: Storage Status ([GET /api/verify/storage-status/:cid](#))

TC-19 — Storage Status Valid CID

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.061
2	Status 200	✓ PASS	0.061
3	cid is present	✓ PASS	0.061
4	status is 'pinned' or 'processing'	✓ PASS	0.061
5	accessible is a boolean	✓ PASS	0.061
6	size is a number	✓ PASS	0.061

TC-20 — Storage Status Unknown CID — expect 404

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.053

#	Assertion	Result	Time (s)
2	Status 404	✓ PASS	0.053
3	Error is 'Report not found'	✓ PASS	0.053

TC-21 — Storage Status No API Key — expect 401 or 403

#	Assertion	Result	Time (s)
1	Response time < 30s	✓ PASS	0.049
2	Status 401 or 403 when no API key	✓ PASS	0.049

Appendix B — Functional Test Suite Results

Suite Name: VERACISM Functional Test Suite **Execution Date:** 2026-03-31 **Total Test Suites (User Journeys):** 19 **Total Assertions:** 89 **Failures:** 0 | **Errors:** 0 **Total Execution Time:** 116.641 seconds

B.1 — F1: IT Office API Tests

Simulates the IT Office staff role uploading documents via the system API using Auth0 Bearer token authentication.

TC-ITO-001 — API Upload Integration

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	21.800
2	TC-ITO-001: Status 200 – upload accepted	✓ PASS	21.800
3	TC-ITO-001: requestId returned	✓ PASS	21.800
4	TC-ITO-001: Results array with at least 1 item	✓ PASS	21.800
5	TC-ITO-001: SHA-256 is 64-char hex	✓ PASS	21.800
6	TC-ITO-001: reportId is a number	✓ PASS	21.800
7	TC-ITO-001: threatLevel is valid enum	✓ PASS	21.800

TC-ITO-002a — Unauthorized Upload — Invalid Bearer Token

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.066
2	TC-ITO-002a: Status 401 for invalid Bearer token	✓ PASS	0.066
3	TC-ITO-002a: No storage occurred	✓ PASS	0.066

TC-ITO-002b — Unauthorized Upload — Missing Authorization Header

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.056
2	TC-ITO-002b: Status 401 when no Authorization header	✓ PASS	0.056
3	TC-ITO-002b: No file stored	✓ PASS	0.056

TC-ITO-003 — Metadata Consistency — Retrieve and Compare

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.131
2	TC-ITO-003: Status 200	✓ PASS	0.131
3	TC-ITO-003: CID matches uploaded CID	✓ PASS	0.131
4	TC-ITO-003: SHA-256 matches	✓ PASS	0.131
5	TC-ITO-003: filename is non-empty	✓ PASS	0.131
6	TC-ITO-003: createdAt is a valid date	✓ PASS	0.131
7	TC-ITO-003: status is complete (2)	✓ PASS	0.131

TC-ITO-004 — Lighthouse Anchoring — CID Verifiable on Public Endpoint

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	10.434
2	TC-ITO-004: Status 200	✓ PASS	10.434
3	TC-ITO-004: found is true (CID exists in VERACISM)	✓ PASS	10.434
4	TC-ITO-004: inLighthouse is true (anchored to Lighthouse)	✓ PASS	10.434
5	TC-ITO-004: sha256Match is true (integrity confirmed)	✓ PASS	10.434
6	TC-ITO-004: CID in response matches uploaded CID	✓ PASS	10.434
7	TC-ITO-004: storedSha256 is 64-char hex	✓ PASS	10.434

B.2 — F2: Registrar Upload Tests

Simulates the Registrar role uploading finalized academic documents with student associations and downloading controlled copies.

TC-REG-001 — Upload Finalized Report

#	Assertion	Result	Time (s)
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#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	17.020
2	TC-REG-001: Status 200 – upload accepted	✓ PASS	17.020
3	TC-REG-001: Tracking ID returned	✓ PASS	17.020
4	TC-REG-001: Results array present	✓ PASS	17.020
5	TC-REG-001: reportId assigned	✓ PASS	17.020
6	TC-REG-001: CID generated (Lighthouse storage)	✓ PASS	17.020

TC-REG-002 — Reject Unsupported File Type

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.061
2	TC-REG-002: Status 400 – validation error	✓ PASS	0.061
3	TC-REG-002: Error message indicates no files	✓ PASS	0.061
4	TC-REG-002: No results array	✓ PASS	0.061

TC-REG-003 — Associate Report to Correct Student

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	17.031
2	TC-REG-003: Status 200	✓ PASS	17.031
3	TC-REG-003: studentId is non-null	✓ PASS	17.031
4	TC-REG-003: Correct student associated	✓ PASS	17.031

TC-REG-007 — Controlled Download — File Returned, Event Logged

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	5.396
2	TC-REG-007: Status 200 – file downloaded	✓ PASS	5.396
3	TC-REG-007: Content-Disposition is attachment	✓ PASS	5.396
4	TC-REG-007: Content-Type header present	✓ PASS	5.396
5	TC-REG-007: Response body is non-empty	✓ PASS	5.396
6	TC-REG-007: No Lighthouse URL in response headers	✓ PASS	5.396

Simulates students and external parties (e.g., employers, other institutions) verifying documents without requiring an account.

TC-STU-002 — Verify by CID Entry — Valid CID

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	7.077
2	TC-STU-002: Status 200	✓ PASS	7.077
3	TC-STU-002: found is true	✓ PASS	7.077
4	TC-STU-002: sha256Match is true (document not tampered)	✓ PASS	7.077
5	TC-STU-002: CID matches submitted CID	✓ PASS	7.077
6	TC-STU-002: storedSha256 is 64-char hex	✓ PASS	7.077
7	TC-STU-002: reportInfo.uploadedAt present	✓ PASS	7.077
8	TC-STU-002: inLighthouse is true	✓ PASS	7.077

TC-STU-003 — Invalid CID Handling — Warning Shown

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.056
2	TC-STU-003: Status 404 for unknown CID	✓ PASS	0.056
3	TC-STU-003: Error message indicates CID not found	✓ PASS	0.056

TC-EXT-001 — External Verify by QR (API Equivalent) — No Auth Required

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	6.629
2	TC-EXT-001: Status 200 without any authentication (public)	✓ PASS	6.629
3	TC-EXT-001: found is true – no login required	✓ PASS	6.629
4	TC-EXT-001: sha256Match is true	✓ PASS	6.629
5	TC-EXT-001: Result includes CID and storedSha256	✓ PASS	6.629

TC-EXT-002 — External Verify by CID — Valid Document

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	6.189
2	TC-EXT-002: Status 200 for valid CID	✓ PASS	6.189

#	Assertion	Result	Time (s)
3	TC-EXT-002: found is true for authentic document	✓ PASS	6.189
4	TC-EXT-002: sha256Match is true (integrity confirmed)	✓ PASS	6.189
5	TC-EXT-002: storedSha256 and CID present	✓ PASS	6.189

TC-EXT-002b — External Verify — Unavailable CID

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.056
2	TC-EXT-002b: Status 404 for unavailable CID	✓ PASS	0.056
3	TC-EXT-002b: Error message indicates CID not found	✓ PASS	0.056

TC-EXT-004 — Restricted Internal Access — Download Without Auth Denied

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.052
2	TC-EXT-004: Direct download without Bearer token returns 401/403	✓ PASS	0.052

TC-EXT-004b — Restricted Internal Access — Reports Endpoint Without Auth Denied

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.054
2	TC-EXT-004b: Reports/my-reports without Bearer token returns 401/403	✓ PASS	0.054

B.4 — F4: End-to-End Journey Tests

Validates the complete upload-to-verification lifecycle in a single automated run, including a tampered-document negative test case.

TC-E2E-001 — Upload-to-Verify Journey (Step 1: Upload)

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	17.744
2	TC-E2E-001 Step 1: Upload succeeds (status 200)	✓ PASS	17.744
3	TC-E2E-001 Step 1: requestId and CID generated	✓ PASS	17.744
4	TC-E2E-001 Step 1: SHA-256 hash computed	✓ PASS	17.744

TC-E2E-001 — Upload-to-Verify Journey (Step 2: External Verifier Checks CID)

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	6.733
2	TC-E2E-001 Step 2: Public verify returns 200	✓ PASS	6.733
3	TC-E2E-001 Step 2: found is true	✓ PASS	6.733
4	TC-E2E-001 Step 2: sha256Match is true	✓ PASS	6.733
5	TC-E2E-001 Step 2: CID matches uploaded CID	✓ PASS	6.733
6	TC-E2E-001 Step 2: computedSha256 matches upload hash	✓ PASS	6.733
7	TC-E2E-001: Full journey complete – upload to verify succeeded	✓ PASS	6.733

TC-E2E-002 — Tampered File Negative Case — Hash Mismatch

#	Assertion	Result	Time (s)
1	[Global] Response time < 30s	✓ PASS	0.056
2	TC-E2E-002: Status 404 – fabricated CID not found in VERACISM	✓ PASS	0.056
3	TC-E2E-002: Error message confirms CID is unknown (tamper indicator)	✓ PASS	0.056

Appendix C — Test Summary and Statistics

C.1 — Integration Test Suite Summary

Group	Test Cases	Assertions	Pass	Fail	Duration (s)
1 — Scan Upload	8	37	37	0	77.969
2 — Get Report by CID	4	17	17	0	0.307
3 — Verify (Public)	4	18	18	0	0.230
4 — Download	3	12	12	0	6.014
5 — Storage Status	3	11	11	0	0.163
TOTAL	22	93	93	0	83.728

C.2 — Functional Test Suite Summary

Feature Group	Test Suites	Assertions	Pass	Fail	Duration (s)
F1 — IT Office API Tests	5	27	27	0	54.231
F2 — Registrar Upload Tests	4	20	20	0	39.508
F3 — Public Verification Tests	7	28	28	0	21.047
F4 — E2E Journey	3	14	14	0	24.533

Feature Group	Test Suites	Assertions	Pass	Fail	Duration (s)
TOTAL	19	89	89	0	116.641

C.3 — Combined Overall Results

Metric	Integration Suite	Functional Suite	Combined
Test Suites / Collections	22	19	41
Total Assertions	93	89	182
Passed	93	89	182
Failed	0	0	0
Errors	0	0	0
Pass Rate	100%	100%	100%
Total Execution Time	83.728 s	116.641 s	200.369 s

C.4 — Key Findings

The following system behaviors were confirmed by automated testing:

1. **Threat Detection Pipeline:** The YARA-based scanner correctly flags files with embedded webshell signatures (TC-07A) and blocks their upload to Lighthouse decentralized storage, confirming that threat detection gates function as designed.
2. **SHA-256 Integrity Verification:** All verification tests confirmed that the SHA-256 hash computed at upload time is correctly stored and matched during public verification, providing tamper-evidence for academic documents.
3. **Lighthouse Decentralized Anchoring:** Uploaded documents receive a CID (Content Identifier) that is verifiable both within VERACISM and directly on the Lighthouse/Filecoin network (TC-ITO-004, TC-EXT-001, TC-EXT-002).
4. **Authentication and Authorization Boundaries:** All protected endpoints correctly reject requests with invalid or missing tokens (401/403). The public verification endpoint requires no authentication, confirming the intended open-access design for external verifiers.
5. **Tenant Isolation:** Report retrieval for a non-owned CID returns 404 rather than 403, preventing tenant enumeration and confirming that tenant data boundaries are enforced without leaking existence information (TC-11).
6. **End-to-End Journey:** The full upload-to-verify workflow (F4/TC-E2E-001) was completed successfully in a single automated run, with the SHA-256 computed at upload matching the hash returned by the public verify endpoint, confirming end-to-end data integrity.
7. **Tampered Document Detection:** A fabricated/unknown CID correctly returns 404, confirming that the system cannot be tricked into verifying documents that were never processed through the platform (TC-E2E-002).

This appendix was generated from automated test results produced by the Newman CLI runner (Postman collection runner). Source files: `tests/results/results.xml` and `tests/results/functional-results.xml`.